Maharshi Karve Stree Shikshan Samstha's
Cummins College of Engineering for Women, Pune
(An autonomous Institute affiliated to Savitribai Phule Pune University)
Department of Computer Engineering

20PECE 601A : DevOps Fundamentals

T1 Evaluation Rubrics

Sem II, 2025-2026

Course Outcomes:

1. Apply the fundamental concepts and emerging trends of DevOps to software development.
2. Analyze the various concepts of application development such as agile software model, microservices to understand the need of DevOps based solution of a system.
3. Compare various concepts and tools of continuous integration, continuous delivery, continuous testing and monitoring for DevOps based realization of solution of a system.
4. Analyze the various deployment platforms as part of DevOps lifecycle.

Project guidelines:

1. A DevOps mini project involves developing a simple application and implementing a complete CI/CD pipeline to automate its lifecycle.
2. The project begins with a clearly defined problem, followed by maintaining the code in a Git repository with proper version control practices.
3. An automated build process is configured using tools like Maven or npm, and Continuous Integration is set up to run tests on every code change.
4. The application is then containerized using Docker to ensure consistency across environments and deployed to a staging environment (local or cloud) for testing.
5. The pipeline configuration, build reports, and deployment evidence are documented to demonstrate a fully functional DevOps workflow.

S.N	Criteria	Excellent	Good	Satisfactory	Needs Improvement
1	Problem Definition & Project Planning (tool usage in each stage of pipeline) (2 Marks) CO3,CO4	Clear and well-structured problem statement with well-defined objectives and relevance (2m)	Problem defined with mostly clear objectives (1.5m)	Basic idea mentioned but limited clarity (1.2)	Problem vaguely defined with minimal explanation (1m)
2	Version Control Usage – Git (4 Marks) CO3	Repository well organized with regular commits, branching strategy, and clear README (4m)	Regular commits with some documentation (3m)	Few commits with minimal explanation (2m)	Repository poorly maintained (1m)
3	Automated Build Process (4 Marks) CO3	Build tool fully integrated and automatically executed in CI pipeline (4m)	Build automation implemented but minor issues (3m)	Build tool used but not fully automated (2m)	Minimal or incorrect build setup (1m)
4	Continuous Integration & Testing (5 Marks) CO3	CI pipeline builds project and runs automated tests with clear reports (5m)	CI pipeline implemented with limited tests (4m)	CI partially configured (3m)	CI pipeline poorly configured (1–2m)
5	Containerization (Docker) (3 Marks) CO4	Application fully containerized with working Dockerfile and image (3m)	Containerization implemented with minor issues (2m)	Partial container setup (1.5m)	Minimal containerization (1m)
6	Deployment to Staging / Cloud (3 Marks) CO4	Application successfully deployed with automated pipeline (3m)	Deployment completed but partially automated (2m)	Manual deployment demonstrated (1.5m)	Deployment not working properly (1m)
7	Demonstration and Question and Answers(4 Marks) CO3,CO4	Clear documentation, architecture explanation, and confident demonstration answering all questions (4m)	Demonstration clear and answers most questions (3m)	Basic documentation and limited explanation during demo (2m)	Very minimal documentation and unclear demonstration (1m)
	Total=25 marks				

Prakash Date
Course Chairman

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